Модуль 4. Практика 3.

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1)

```
Switch#show vla
Switch#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi2/0
20   VLAN20                           active
100  VLAN100                          active
200  VLAN0200                         active
300  VLAN0300                         active
333  VLAN333                          active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
```

VLAN 20 и 333 существуют, но пусты.

Настройка интерфейсов на первом маршрутизаторе

```
Switch(config)#int range gi0/3,gi1/1,gi1/3
Switch(config-if-range)#switchport trunk enc dot1
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#switchport trunk allow vlan 333

Switch(config)#int range gi0/2,gi1/0,gi1/2
Switch(config-if-range)#switchport trunk enc dot1
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk allow vlan 20
```

show int trunk

```
Switch#show int trunk

Port        Mode             Encapsulation  Status        Native vlan
Gi0/0       desirable        n-isl          trunking      1
Gi0/1       desirable        n-isl          trunking      1
Gi0/2       on               802.1q         trunking      1
Gi0/3       on               802.1q         trunking      333
Gi1/0       on               802.1q         trunking      1
Gi1/1       on               802.1q         trunking      333
Gi1/2       on               802.1q         trunking      1
Gi1/3       on               802.1q         trunking      333

Port        Vlans allowed on trunk
Gi0/0       1-4094
Gi0/1       1-4094
Gi0/2       20
Gi0/3       333
Gi1/0       20
Gi1/1       333
Gi1/2       20
Gi1/3       333

Port        Vlans allowed and active in management domain
Gi0/0       1,20,100,200,300,333

Port        Vlans allowed and active in management domain
Gi0/1       1,20,100,200,300,333
Gi0/2       20
Gi0/3       333
Gi1/0       20
Gi1/1       333
Gi1/2       20
Gi1/3       333

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/0       1,20,100,200,300,333
Gi0/1       20,100,200,300,333
Gi0/2       20
Gi0/3       333
Gi1/0       20
Gi1/1       333
Gi1/2       20
Gi1/3       333
Switch#
```

Настройка интерфейсов на коммутаторах 3-5

```
Switch(config)#int range g0/0-1
Switch(config-if-range)#switchport trunk enc dot1
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#exit
Switch(config)#int g0/0
Switch(config-if)#switchport trunk allow vlan 20
Switch(config-if)#exit
Switch(config)#int g0/1
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#switchport trunk allow vlan 333
Switch(config-if)#exit
Switch(config)#int g1/0
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int g1/1
Switch(config-if)#switchport access vlan 333
```

sh vlan br

sh int tr

```
% Type "show ?" for a list of subcommands
Switch#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
20   VLAN20                           active    Gi1/0
100  VLAN100                          active
200  VLAN0200                         active
300  VLAN0300                         active
333  VLAN333                          active    Gi1/1
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
Switch#
Switch#show int tr
Switch#show int trunk

Port        Mode             Encapsulation  Status        Native vlan
Gi0/0       on               802.1q         trunking      1
Gi0/1       on               802.1q         trunking      333
Gi0/2       desirable        n-isl          trunking      1
Gi0/3       desirable        n-isl          trunking      1

Port        Vlans allowed on trunk
Gi0/0       20
Gi0/1       333
Gi0/2       1-4094
Gi0/3       1-4094

Port        Vlans allowed and active in management domain
Gi0/0       20
Gi0/1       333
Gi0/2       1,20,100,200,300,333
Gi0/3       1,20,100,200,300,333

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/0       20
Gi0/1       333
Gi0/2       1,20,100,200,300,333
Gi0/3       20,100,200,300,333
Switch#
```

2)

```
PC1> ip 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> ping 192.168.1.2

host (192.168.1.2) not reachable

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=6.233 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=4.497 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=3.526 ms
^C
PC1>  ping 192.168.1.4

host (192.168.1.4) not reachable

PC1>  ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=4.043 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=9.889 ms
^C
PC1>  ping 192.168.1.6

host (192.168.1.6) not reachable

PC1> save
Saving startup configuration to startup.vpc
.  done
```

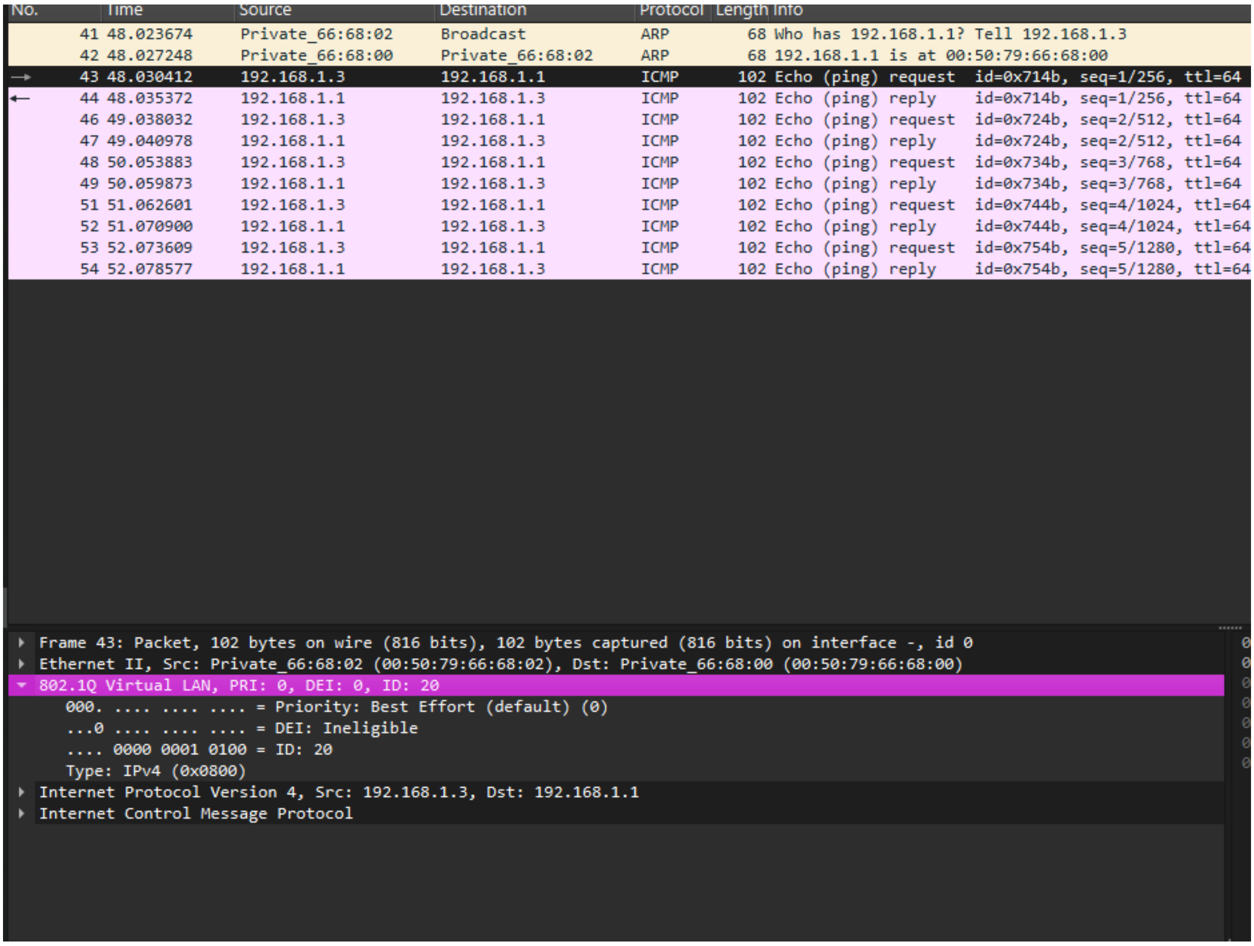
```
PC2> ip 192.168.1.2
Checking for duplicate address...
PC2 : 192.168.1.2 255.255.255.0

PC2>  ping 192.168.1.1

host (192.168.1.1) not reachable

PC2>  ping 192.168.1.3

host (192.168.1.3) not reachable

PC2>  ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=2.214 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=2.129 ms
^C
PC2>  ping 192.168.1.5

host (192.168.1.5) not reachable

PC2>  ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=5.474 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=5.711 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=4.991 ms
^C
PC2> save
Saving startup configuration to startup.vpc
.  done

PC2> 
```

```
PC3>  ip 192.168.1.3
Checking for duplicate address...
PC3 : 192.168.1.3 255.255.255.0

PC3>  ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=7.039 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=12.613 ms
^C
PC3>  ping 192.168.1.2

host (192.168.1.2) not reachable

PC3> save
Saving startup configuration to startup.vpc
.  done

PC3>
PC3> ping 192.168.1.4

host (192.168.1.4) not reachable

PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=13.983 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=8.586 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=15.900 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=2.447 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=4.794 ms
^C
PC3> ping 192.168.1.6

host (192.168.1.6) not reachable
```

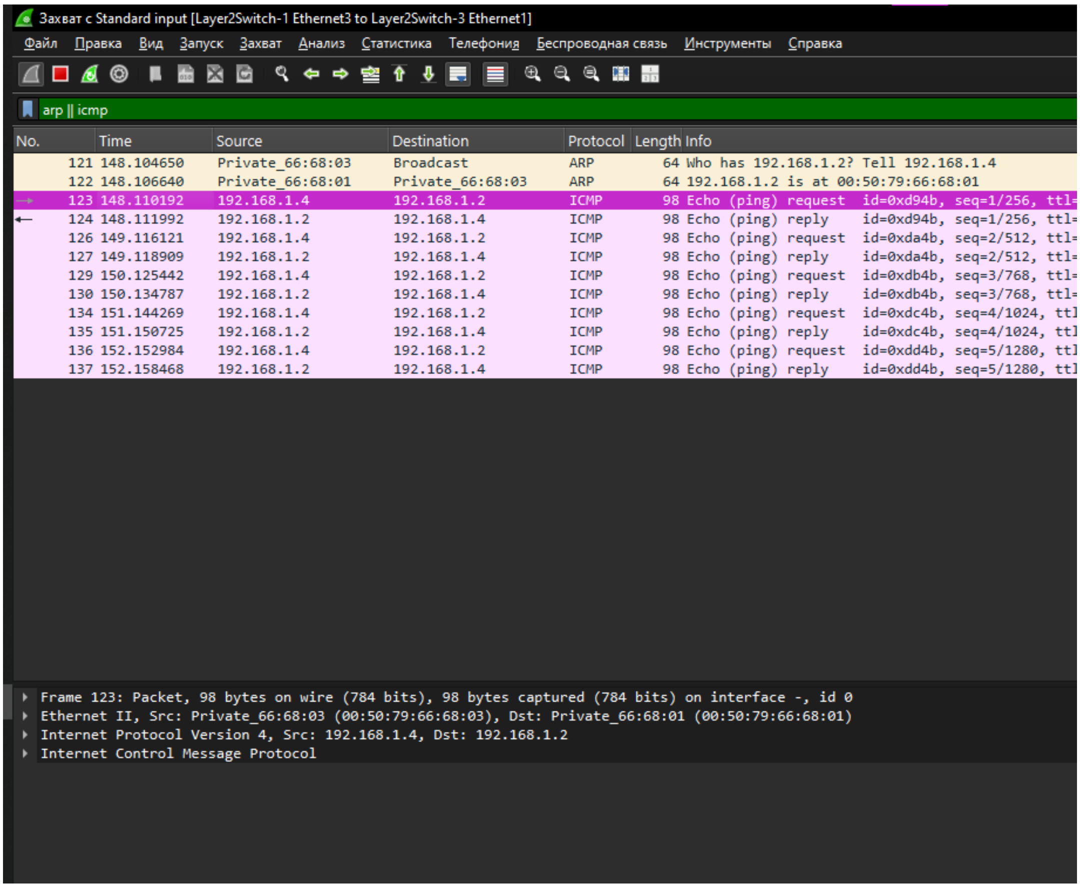
3)

Теги содержатся в пакетах передаваемых в VLAN 20

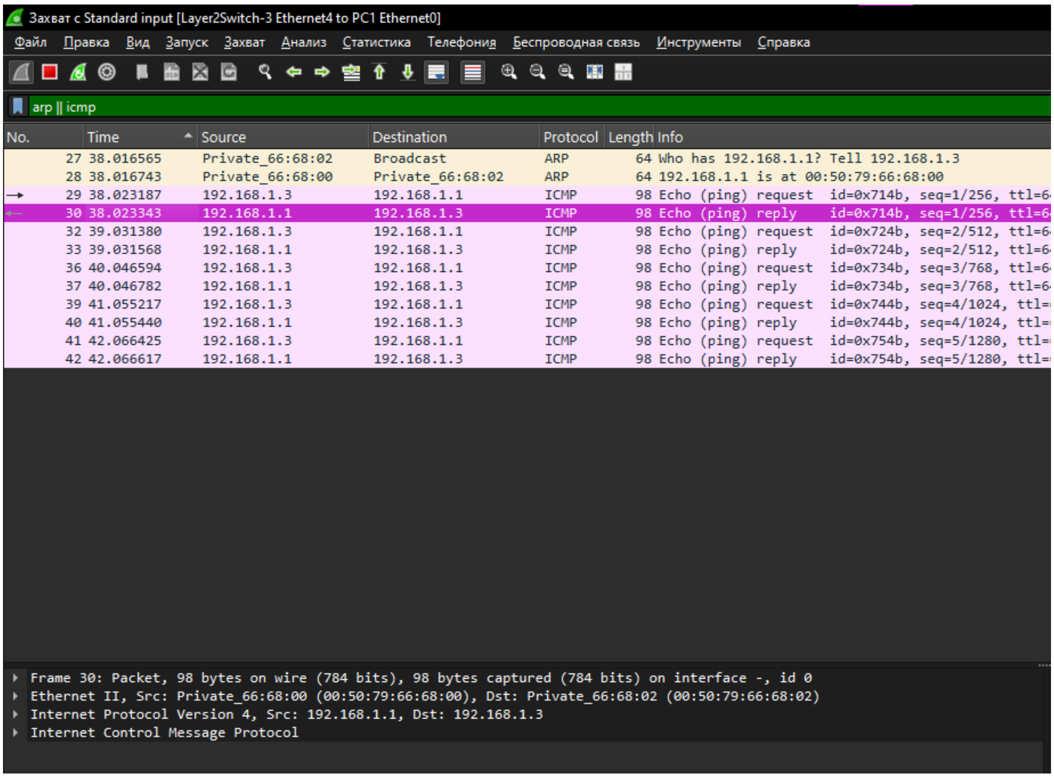
(отслеживается trunk линк, сам тег ввыделен)



Тегов не будет в случае access портов и native trunk:



Native trunk



access порт

4)